Multiexpert Automatic Speech Recognition Using Acoustic and Myoelectric Signals

Adrian D. C. Chan*, *Senior Member, IEEE*, Kevin B. Englehart, *Senior Member, IEEE*, Bernard Hudgins, *Senior Member, IEEE*, and Dennis F. Lovely, *Member, IEEE*

***Abstract*—Classification accuracy of conventional automatic speech recognition (ASR) systems can decrease dramatically under acoustically noisy conditions. To improve classification accuracy and increase system robustness a multiexpert ASR system is implemented. In this system, acoustic speech information is supplemented with information from facial myoelectric signals (MES). A new method of combining experts, known as the plausibility method, is employed to combine an acoustic ASR expert and a MES ASR expert. The plausibility method of combining multiple experts, which is based on the mathematical framework of evidence theory, is compared to the Borda count and score-based methods of combination.**

Acoustic and facial MES data were collected from 5 subjects, using a 10-word vocabulary across an 18-dB range of acoustic noise. As expected the performance of an acoustic expert decreases with increasing acoustic noise; classification accuracies of the acoustic ASR expert are as low as 11.5%. The effect of noise is significantly reduced with the addition of the MES ASR expert. Classification accuracies remain above 78.8% across the 18-dB range of acoustic noise, when the plausibility method is used to combine the opinions of multiple experts. In addition, the plausibility method produced classification accuracies higher than any individual expert at all noise levels, as well as the highest classification accuracies, except at the 9-dB noise level. Using the Borda count and score-based multiexpert systems, classification accuracies are improved relative to the acoustic ASR expert but are as low as 51.5% and 59.5%, respectively.

***Index Terms*—Classification, electromyography, multiexpert system, myoelectric signals, speech recognition.**

I. Introduction

AUTOMATIC speech recognition (ASR) is being investigated as an alternative control methodology for pilots flying high performance jet aircraft [1]. ASR has the potential of increasing pilot efficiency by simplifying the control interface, and improving pilot safety by encouraging "heads-up" flying; however, the performance of conventional acoustic ASR systems can be severely degraded by noise conditions during flight (e.g., ambient cockpit noise, high G-force, vibration). Although research is being conducted to improve acoustic ASR under noisy conditions [2], [3], it is expected that the performance of any mono-modal approach will eventually saturate in performance [4].

In 2001, the authors proposed the use of facial myoelectric signals (MES) as a second source of speech information and demonstrated the potential of MES ASR [5]. This preliminary work employed a classification methodology used for myoelectrically controlled powered prostheses, which employed linear discriminant analysis (LDA) on a set of wavelet transform features that were reduced by principal component analysis [6]. A large improvement in classification accuracy was found in comparison to previous work in MES ASR [7]–[10]. The authors later developed new classification methodology for MES ASR based on hidden Markov models (HMMs) [11]. A HMM is a double embedded stochastic process that consists of a Markov chain topology and probability density functions associated with each state, which govern the observable output. This structure enables a HMM to simultaneously characterize both local signal properties (i.e., regularities that occur within a short time span) and global signal properties (i.e., dynamic relationship among local signal properties). The majority of conventional acoustic ASR systems use HMMs to classify the acoustic information. It was felt that HMMs would also be well suited for MES ASR as the articulation of words could be thought of as ordered sequence of events. In addition, the structure of HMMs enable them to cope with time-scale and shape variances associated with speech. Indeed, the HMM classifier was found to be more robust than the LDA classifier [11]. MES ASR has also been successfully implemented using artificial neural networks (ANNs) [12], [13]; however, it is expected that the performance of an ANN classifier would be susceptible to temporal variations in much the same way as the LDA classifier.

Previous work on MES ASR has demonstrated the feasibility of a multimodal ASR system using acoustic and MES information. It was thought that an MES ASR expert would effectively complement an acoustic ASR expert for two main reasons. First, MES should not be affected by acoustic noise. Second, there exists word pairs that sound acoustically similar but are articulated in a dissimilar manner, which would manifest distinctively in the MES (e.g., "sign" and "fine"). While an acoustic ASR expert may have difficulties discerning these acoustically similar words, a MES ASR expert could accomplish this task with relative ease. This paper presents a new method of combining the opinion of multiple experts and demonstrates the effectiveness of this multimodal ASR approach under acoustically noisy conditions.

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II. Background

A. Single Expert System

The general task of pattern recognition can be represented mathematically by first defining a pattern space or frame of discernment Θ, which is the M mutually exclusive sets or classes, denoted C_i ($i \in \Lambda = \{1, 2, \ldots, M\}$). That is

$$\Theta = \bigcup_{i=1}^{M} C_i. \tag{1}$$

The task of pattern recognition is to map an unknown signal $x \in \Theta$ to a particular class C_i. Pattern recognition is employed in a variety of applications, such as ASR, handwriting recognition, and person identification. Pattern recognition can be implemented using an expert, which contains two functional blocks: a feature extractor and a classifier. The process of feature extraction transforms an input signal into a finite set of features that will be used to discern the signal. For example, in person identification the input signal may be a picture of a person and the set of features may include hair color, height, and weight. These features form a feature vector that is used as the input to the classifier and the classifier provides the mapping from the feature vector to a particular class. The mapping relationship is typically derived from a training set of data, which consists of sample inputs from known classes.

B. Multiexpert System

Having multiple input signals can be an effective method of improving classification accuracy and system robustness over varying conditions. Humans use a form of multimodal speech recognition that uses both acoustic and visual inputs. Research has shown that speech intelligibility improves when both of these sources of information are used, especially under acoustically noisy conditions [14], [15]. Using acoustic and visual inputs have also successfully increased the classification accuracy for automatic person identification systems, compared to systems that only use one of these inputs [4], [16]. In this paper, we demonstrate that using MES inputs can be used to improve the performance of a conventional acoustic ASR system.

When multiple sources of information are used for pattern recognition, it is possible to utilize this information within a signal expert; however, it is often preferable to use parallel experts that are in series with a supervisor or classifier, which is used to make the final classification decision. In [17], three main reasons are cited in support of a multiexpert approach.

1) The type of information from each source may be different, requiring features that are drastically different; this could make it difficult to amalgamate these features into a single classifier. For example, features that are continuous in time may be difficult to combine with discrete time features or static features.
2) Features may differ significantly in what they represent, making it difficult to accommodate them in a single classifier. For example, in acoustic ASR, if one set of features was speech volume and another set of features was mouth opening, it would be difficult to lump these features into a single classifier. The task of appropriately scaling the two features could be a difficult task in itself. With a multiexpert system, information sources are combined using the expert outputs, which have similar properties (e.g., confidence values); combination of the opinions of experts can be performed with relative ease.
3) Additional sources of information imply additional features; this can result in a feature vector with high dimensionality for a single classifier. A high dimensionality classification problem can significantly increase the computational complexity, causing accuracy problems and typically requires a larger set of training examples.

C. Combining the Opinion of Experts

The performance of a multiexpert system is not only dependent on individual experts but also on the method used to combine the opinions of the experts. Before describing methods of combining the opinions of experts, it is first necessary to expand the mathematical description of pattern recognition for a multiexpert system. Suppose there are K experts, denoted e_k ($k = \{1, 2, \ldots, K\}$) and that we are given an unknown pattern $x \in \Theta$ to classify. In general, an expert can provide an output at one of three levels [16], [17].

1) *Abstract Level:* An expert e_k assigns the unknown pattern x an index $i \in \Lambda$, which indicates the expert's opinion that $x \in C_i$. Note that some experts also allow the assignment of the index $M + 1$, which is used to indicate that the unknown pattern x is rejected; that is, expert e_k cannot discern, with sufficient certainty, to which class the unknown pattern x belongs.
2) *Rank Level:* An expert e_k assigns the unknown pattern x a rank vector $V_k = [v_k(1)\ v_k(2)\ \ldots\ v_k(M)]^T$, which is a permutation of the set Λ. Elements of the rank vector $v_k(j)$ are ordered in descending order according to the confidence that the expert considers $x \in C_i$, where $i = v_k(j)$. Therefore, expert e_k associates the highest degree of confidence with the proposition $x \in C_i$, where $i = v_k(1)$.
3) *Measurement Level:* An expert e_k assigns the unknown pattern x a measurement vector $S_k = [s_k(1)\ s_k(2)\ \ldots\ s_k(M)]^T$. Elements of the measurement vector $s_k(i)$ provide an indication of the degree of the confidence that an expert associates with the proposition $x \in C_i$. The higher the value of $s_k(i)$, the higher the associated degree of confidence.

Of these three levels of expert output, the measurement level contains the most information (highest level), while the abstract contains the least (lowest level). An expert's output can always be reduced from a higher level to a lower level. For example, an output on the measurement level can be reduced to the rank level by using the measurement vector S_k to order a permutation of Λ, creating the rank vector V_k. A rank level output can be reduced to the abstract level by discarding everything in the rank vector, except the first element $v_k(1)$.

The objective of combining the opinion of experts is to form a multiexpert decision $E(x) = C_i$, based on the consolidated information from the individual experts. When combining the opinion of experts, it is necessary to ensure that the experts

provide outputs on the same level. This implies that methods that combine the opinion of experts on the abstract level are the most generalizable methods, as all expert outputs an be reduced to this level; however, a method that operates on a higher level should have better performance as it utilizes more information. Thus, there exists a tradeoff between generalizability and performance.

In this paper, we present a new method of combining the opinion of experts, which we will refer to as the plausibility method. The performance of the plausibility method will be evaluated against two conventional methods of combining the opinions of experts: the Borda count method and a score based method.

D. Borda Count

The Borda count method of combining the opinion of experts [18] can be considered a generalization of a majority vote method. This method is a common method of combination, which combines the opinion of experts at the rank level.

The Borda count $\mathrm{r_k(i)}$ is a score that expert $\mathrm{e_k}$ associates with class $\mathrm{C_i}$. The Borda count is computed using the rank vector $\mathrm{V_k}$ as follows:

$$\mathrm{r_k(i) = M - \arg_m [v_k(m) = i]}. \tag{2}$$

The higher a class is ranked in the rank vector, the larger the Borda count, with the lowest ranked class receiving a Borda count of zero. Borda counts are consolidated to form a multiexpert decision by simply summing the Borda counts across the individual experts

$$\mathrm{R(i)} = \sum_{\mathrm{k=1}}^{\mathrm{K}} \mathrm{r_k(i)}. \tag{3}$$

The output of the multiexpert system is a rank level output, with rank vector $\mathrm{V_E} = [\mathrm{v_E(1)\ v_E(2)\ \ldots\ v_E(M)}]^{\mathrm{T}}$, which is a permutation of set Λ, ordered by descending Borda count R(i). Classes with equal Borda counts can be handled in a variety of methods, such as randomly settling ties, or biasing decisions to a particular expert.

E. Score-Based Methods

Score-based methods of combining the opinion of experts operate at the measurement level. A transformation of the measurements vectors or score vectors $\mathrm{S_k} = [\mathrm{s_k(1)\ s_k(2)\ ,\ldots,\ s_k(M)}]^{\mathrm{T}}$ into normalized scores is required so that the range of scores is the same across all experts. In [19], four such transformations are described: linear, logarithmic, exponential and logistic.

The various nonlinear transformations (i.e., logarithmic, exponential, and logistic) are applied to emphasize or de-emphasize portions of the scoring range. In both [19] and [20], it was found that the logarithmic transformation yielded the best results on average and was used in this paper.

Once the scores have been normalized, they can be combined using relatively basic mathematical operations. In [20], four such methods are described: sum, difference, weight, and median.

Experimental results have shown the weight method of combination yielded the best overall classification accuracy among the four methods listed above [20]; however, this result was not consistent across all test sets. In addition, there was only a small difference in the classification accuracies between the four methods of combination. The sum method is used in this paper since the weight method requires additional training for proper selection of the weights and the expected difference classification accuracy is minimal.

III. Plausibility Method

The plausibility method is a new method of combining the opinion of experts that operates on the measurement level. This method is based on the mathematical theory of evidence, which was first introduced by Dempster [21] and later refined by Shafer [22]. Evidence theory forms a mathematical framework that enables precise assignment of partial beliefs and a method of combining partial beliefs from multiple bodies of evidence.

A. Evidence Theory

Evidence theory uses the frame of discernment Θ, which was defined in (1). The power set of Θ is denoted 2^{Θ} and refers to the set of all subsets of Θ, including the whole set Θ and empty set $\varnothing$. In evidence theory, we concern ourselves with propositions of the form "the correct class is set A", where $\mathrm{A} \in 2^{\Theta}$.

The basic probability assignment (BPA) m(A) is a function that is used to precisely assign a portion of belief to a set $\mathrm{A} \in 2^{\Theta}$. By precisely, we mean that this assignment does not imply any additional claims regarding the portions of belief that are assigned to the negation of A or to any subset of A[1]. A BPA has the following constraints:

$$1) \quad \mathrm{m(A)} \in [0,1] \tag{4}$$

$$2) \quad \mathrm{m}(\varnothing) = 0 \tag{5}$$

$$3) \quad \sum_{\mathrm{A} \subseteq \Theta} \mathrm{m(A)} = 1. \tag{6}$$

The first constraint is that the range of the BPA is between zero and one. The second constraint is that no portion of belief can be assigned to the empty set. The final constraint is that the sum of all the assigned portions of belief must equal one, which is done by convention.

Partial beliefs from two independent bodies of evidence with BPAs $\mathrm{m_1}$ and $\mathrm{m_2}$ can be combined using Dempster's rule of combination:

$$\mathrm{m_{1,2}(A)} = \begin{cases} \dfrac{\sum_{\mathrm{B \cap C = A}} \mathrm{m_1(B) m_2(C)}}{1 - \sum_{\mathrm{B \cap C = \varnothing}} \mathrm{m_1(B) m_2(C)}}, & \text{if } \mathrm{A} \neq \varnothing \\ 0, & \text{if } \mathrm{A} = \varnothing \end{cases}. \tag{7}$$

We can interpret the numerator of (7), for $\mathrm{A} \neq \varnothing$, as follows: if the first body of evidence assigns a portion of belief $\mathrm{m_1(B)}$ to proposition B and the second body of evidence assigns a portion of belief $\mathrm{m_2(C)}$ to proposition C, these beliefs are combined by taking the product $\mathrm{m_1(B) m_2(C)}$ and assigning it to the intersection of the two propositions $\mathrm{A = B \cap C}$. If the propositions B and C do not overlap then their intersection is the empty set

[1]For example, in conventional Bayesian probability theory, the assignment of a probability Pr(A) would imply $\mathrm{Pr(\neg A) = 1 - Pr(A)}$

$\varnothing$. Since no portion of belief can be assigned to the empty set, $m_{1,2}(A)$ is explicitly assigned a value of 0 for $A = \varnothing$. The denominator of (7), for $A \neq \varnothing$, is a scaling factor that ensure that the BPA sums to one [refer to (6)] by accounting for portions of belief that would have been assigned to intersections that equate to the empty set.

If there are more than two bodies of evidence to combine, this is accomplished by iteratively applying Dempster's rule of combination. Dempster's rule of combination is a commutative and associative operation; therefore, the order in which the multiple bodies of evidence are combined does not affect the result.

From the BPA, evidence theory provides two dual nonadditive measures: belief and plausibility. The belief and plausibility measure are the lower and upper bounds of probability for proposition A, respectively. The belief measure Bel(A) is the sum of all portions of belief that are assigned to the set A and all subsets of A; in other words, it is the total belief assuming that all uncertainties do not support proposition A. The plausibility measure Pl(A) is the sum of all portions of belief that assigned to sets that intersect A; in other words, it is the total belief assuming that all uncertainties support proposition A. Mathematically, the belief and plausibility measures are calculated as follows:

$$\mathrm{Bel}(A) = \sum_{B|B\subseteq A} m(B) \tag{8}$$

$$\mathrm{Pl}(A) = \sum_{B|B\cap A\neq\varnothing} m(B) \tag{9}$$

B. Plausibility Method of Combination

Previous research has demonstrated that evidence theory provides a solid foundation for combining the opinion of experts; however, results are highly dependent upon the method used to perform the BPA.

In [17], the combination is performed at the abstract level with the BPA computed using the recognition and substitution rates. Combination at the abstract level, as explained earlier, is undesirable due to a reduction of information from the measurement level. This method is also highly sensitive to the accuracy of the estimated recognition and substitution rates. In [23], estimations of the intra- and inter-class distances are used to calculate class conditional probabilities, which are then transformed into a BPA. As explained in [24], this method brought forward questions regarding the choice of reference vectors and distance vectors. In [24], the BPA is performed by computing the distance between a reference vector and an expert's measurement vector, which are later transformed into a BPA; however, the method of computing the reference vector is not optimal, and also opens questions regarding the method by which the proximity measure were transformed into partial beliefs [25]. In [25], the reference vector is optimized by minimizing the mean square error of the training set output; however, this method requires a large amount of training data to adequately estimate the reference vector.

The plausibility method defines a new procedure for performing the BPA. This method operates on the measurement level and does not base the BPA upon distances from a fixed reference vector. Instead, this procedure was constructed for a dynamic environment where the performance of an individual expert may vary. The BPA is performed by comparing an expert's output to the expert's outputs for its training data. If the expert is properly trained, its output for the training data would represent outputs with a high degree of confidence. Comparing new outputs to the training data outputs will, therefore, provide an indication of the output's reliability. To formulate the BPA the following assumptions are used.

1) Strong supporting evidence appears as a high relative score, which is not necessarily a high absolute score. This implies that an expert that cannot effectively discern an input will produce scores that are similar for all classes. In addition, scores may be shifted in the absolute sense (e.g., due to a single corrupted feature) but this should not necessarily reduce the reliability of this expert's decision, as long as the relative score difference is high.
2) Strong discernment is not associated with two scores that are both very low or two scores that are both very high. Differences in scores are more significant if they fall toward the middle of the expert's output range.
3) Scores associated with misclassifications can be associated with output noise; therefore, strong supporting evidence must exceed this noise

To incorporate these assumptions, the measurement vector S_k generated by expert e_k is not used directly; instead, the zero-mean scores $z_k(i) = s_k(i) - \mathrm{mean}(S_k)$ are used. A nonlinear mapping is also applied to these zero-mean scores, such that is assigns small differences between two scores that are both very low or both very high, enhancing the midrange scores. The nonlinear mapping should be a nondecreasing function, with a range from 0 to 1, so that the transformed scores conform to the definition of a BPA. Although many different nonlinear transformations can be used, a Gaussian cumulative distribution function was employed for simplicity.

The Gaussian cumulative distribution function can be defined entirely by its mean and variance. To determine the values of these parameters, scores generated from the training data were used; specifically, the scores associated with the incorrect classes were used to generate an estimate of the classification noise of each expert. The average variance σ_k^2 of these scores is used to form the Gaussian cumulative distribution function G_k, with mean $2\sigma_k$ and variance σ_k^2. Selection of this mean and variance ensures that small portions of belief are assigned for low scores within the output noise range. With these values for the mean and variance 95% of the assigned belief exists between the mean score and four standard deviations above the mean score. Empirical testing also demonstrated that this parameter selection is close to being optimal.

To classify an unknown signal $x \in \Theta$, the mean of the measurement vector S_k from expert e_k is removed to form the zero-mean score vector $Z_k = [z_k(1)\ z_k(2)\ \ldots\ z_k(M)]$. Using the cumulative distribution function G_k, portions of belief are assigned according to the rule

$$m_k\left[A_k(p)\right] = G_k\left[z_k\left(v_k(p)\right)\right] - G_k\left[z_k\left(v_k(p+1)\right)\right] \tag{10}$$

where

$$A_k(p) = \left\{C_i : i \in \left\{v_k(q) : 1 \leq q < p\right\}\right\}. \tag{11}$$

The set $A_k(p)$ is the union of all classes with a score greater than or equal to the pth highest score (i.e., the first p entries

in the rank vector V_k). Portion of belief assigned to $A_k(p)$ is equal to the difference in the cumulative distribution function, evaluated at the pth highest score and the $(p+1)^{th}$ highest score. Any remaining portions of belief are assigned to the whole set Θ, representing partial beliefs that do not support any specific proposition.

An example of the BPA is depicted in Fig. 1 for a five class problem with $Z_k = [0.5\ -2.0\ 2.0\ 1.0\ -1.5]^T$ and $\sigma_k^2 = 1$; this variance of $\sigma_k^2 = 1$ would have been obtained from the training scores associated with incorrect classes. From this variance, the nonlinear score mapping of the Gaussian cumulative distribution function G_k is constructed with a variance of 1 and a mean of 2, as depicted in the bottom half of Fig. 1. In this example, class C_2 has the lowest score of -2.0, so the mapped score $G_k(-2.0) = 3.2 \times 10^{-5}$ is the partial belief assigned to the set of classes that have a score greater than or equal to the C_2 score, which are all the classes; therefore, this portion of belief is assigned to the whole set Θ. Class C_5 has the next lowest score of -1.5, so the difference in the mapped scores $G_k(-2.0) - G_k(-1.5) = 2.0 \times 10^{-4}$ is the partial belief assigned to set of classes that have a score greater than or equal to the C_5 score, which are all the classes except C_2. This process is continued until class C_3, which has the highest score of 2.0, is exclusively assigned the partial belief of $G_k(2.0) - G_k(1.0) = 0.34$. The remaining unassigned partial belief $G_k(\infty) - G_k(2.0) = 0.5$ is not associated with any particular class and is, therefore, assigned to the whole set Θ (vacuous belief).

Once the BPA has been completed for each expert, the opinions from multiple experts are combined using Dempster's rule of combination, forming a combined BPA m_E. The plausibility measure is computed from m_E and the multiexpert system makes the classification decision based on the highest plausibility. That is

$$E(x) = C_i \quad i = \arg_j \left(\max\left(Pl(C_j)\right)\right). \tag{12}$$

The nonlinear BPA of the scores has some intuitively pleasing results. First, the difference in plausibility between two classes is proportional to the difference in their scores, with the nonlinearity ensuring that differences in very low or very high scores are deemphasized. For example, two classes that both have relatively high scores will be assigned strong evidentiary support relative to the other classes but will have similar evidentiary support compared to each other. Second, it is expected that experts with unreliable outputs will typically have less variability across classes (i.e., classes will have similar scores). The BPA will result in an assignment that is close to the vacuous belief function (i.e .$m(\Theta) = 1$), which is the proper representation of complete uncertainty[2].

IV. Methodology

In this study, a multimodal ASR, using an acoustic ASR expert and a MES ASR expert, is evaluated under a large range of acoustic noise. Performance of the plausibility method is compared to the Borda count method and score-based method.

[2]In conventional Bayesian probability theory, complete uncertainty is represented with equal probabilities (i.e., $Pr(C_i) = (1/M), i = 1, 2, \ldots, M$)

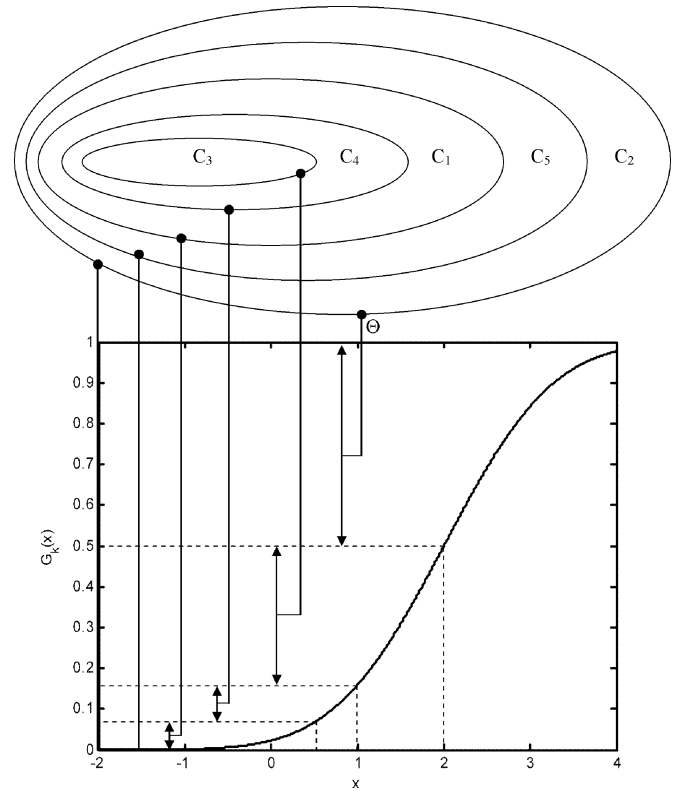

Fig. 1. Basic probability assignment using the plausiblity method for a five class case (C_i, $i = 1, 2, \ldots, 5$) with a zero mean score vector $Z_k = [0.5\ -2.0\ 2.0\ 1.0\ -1.5]^T$ and expert classification noise variance $\sigma_k^2 = 1$.

A. Data Collection

Five Canadian-English speaking male subjects with no known speech disorders participated in this study. A 10-word vocabulary was used, consisting of the digits "zero" to "nine". For each subject, 13 sets of 40 words were constructed, with each set containing four repetitions of each word in the vocabulary (total of 520 words per subject). The order of the words were randomly permuted and visually presented to the subject one at a time, with a 1-s pause between words to minimize coarticulatory and anticipatory effects.

During each set of words, acoustic additive white Gaussian noise was generated using computer speakers. Six different levels of relative noise power were used (0, 6, 9, 12, 15, and 18 dB). The noise power was referenced to the speech power for each individual subject by adjusting the noise power so that the measured amplitude of the highest noise level (18 dB) was equal to the measured peak amplitude of the acoustic speech. At this 18-dB noise level, even human perception of speech was quite difficult. Once this 18-dB level was calibrated, the noise power was adjusted by changing the variance of the noise appropriately (this means that the 0-dB noise level is not noise free, rather has 18-dB less power than the highest noise level). Of the 13 sets of words, the first, seventh, and last set used the 0-dB noise level. Each of the remaining noise levels was used twice and the order randomly permuted among the remaining 10 sessions.

Surface MES data were obtained from five sites of the face (Fig. 2) using Ag-AgCl Duo-Trode electrodes (Myotronics Inc., 6140), with an Ag-AgCl RedDot electrode (3M, 2259)

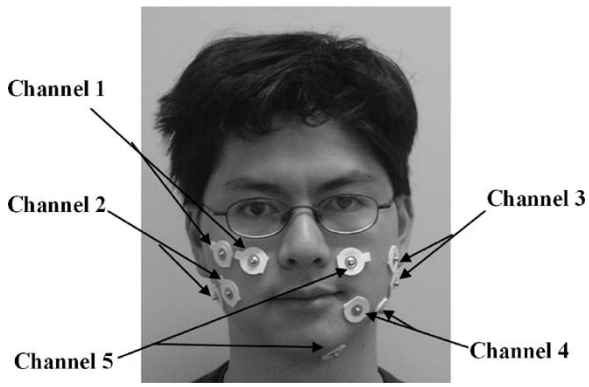

Fig. 2. Electrode placement.

placed on the back of the neck to provide a common reference. MES were preamplified using optically isolated amplifiers [26], bandlimited between 0.1 and 500 Hz, and simultaneously sampled at 10 kHz per channel, using a 12-bit analog-to-digital converter (ComputerBoards, CIO-DAS16/330) and companion sample-and-hold unit (ComputerBoards, CIO-SSH16). MES data were obtained simultaneously with the acoustic signal, which was bandlimited between 0.1 and 5 kHz. Eighth-order Butterworth antialiasing filters were used prior to sampling, with cutoff frequencies of 500 and 5 kHz for the MES and acoustic signal, respectively.

B. Training and Test Sets

Acoustic and MES ASR experts were trained using data from the three 0-dB noise level word sets (12 training examples per word) and tested on the data from the remaining noise level word sets (8 test examples per word, per noise level). In order to test at the 0-dB noise level, a leave-one-out method was used on the 0-dB noise level word sets. In this leave-one-out method, experts were trained using two of the 0-dB noise level word sets and tested on the remaining 0-dB noise level word set. This method was repeated so that all three 0-dB noise level word sets served as the test set. For the 0-dB noise level, there were only 8 training examples per word but a total of 12 test examples per word.

C. Experts

The acoustic expert was a 12-state, left-right HMM, with single mixture Gaussian observation densities. Overlapping observation windows of 25.6 ms were used, with a spacing of 12.8 ms. For each observation window, the first 12 mel-frequency cepstral coefficients, were used as signal features.

The MES expert was a 12-state, left-right HMM, with single mixture Gaussian observation densities. Overlapping observation windows of 128 ms were used, with a spacing of 16 ms. For each observation window, the first 12 mel-frequency cepstral coefficients and the root-mean-squared value were used as signal features.

Likelihoods computed by the HMM were normalized by the number of transition and observation probabilities used to compute the likelihood. These normalized likelihoods were used as the scores to form the measurement vector output for each expert.

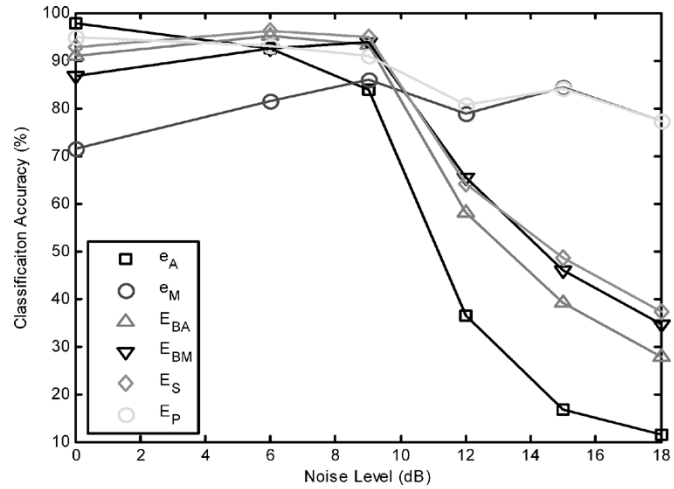

Fig. 3. Classification accuracies for individual experts: acoustic expert $(\mathrm{e_A})$, MES expert $(\mathrm{e_M})$; and two-expert systems: Borda method favoring acoustic expert $(\mathrm{E_{BA}})$, Borda method favoring MES expert $(\mathrm{E_{BA}})$, score-based method $(\mathrm{E_S})$, and plausibility method $(\mathrm{E_P})$.

D. Combining the Opinion of Experts

Classification accuracies were obtained for the acoustic expert $(\mathrm{e_A})$ and the MES expert $(\mathrm{e_M})$. Classification accuracies were also obtained for a multiexpert system combining the opinion of experts using the following.

1) Borda count method favoring the acoustic expert during ties $(\mathrm{E_{BA}})$.
2) Borda count method favoring the MES expert during ties $(\mathrm{E_{BM}})$.
3) Score-based method, using a logarithmic normalization procedure prior to combining them $(\mathrm{E_S})$.
4) plausibility method $(\mathrm{E_P})$

V. Results

The classification results, averaged over the five subjects are shown in Fig. 3. For the acoustic expert, classification accuracy is 97.7% at the 0-dB noise level but a severe decrease in classification accuracy occurs with increasing noise level. At 18 dB, classification accuracies for the acoustic expert are 11.5%, which is near the *a priori* classification accuracy (i.e., $\text{accuracy of randomly guessing} = 10\%$). Note that the classification accuracy at the 0-dB level is likely lower than expected due to the reduced number of training examples at this level.

Classification accuracy of the MES expert shows no definite trend with increasing noise power. Any possible effects of noise on articulation, such as the Lombard effect,[3] are either negligible or masked by the variability of the MES expert. An average classification accuracy of 79.9% is computed across all subjects and noise levels. As with the acoustic expert, classification accuracy of the MES expert at 0-dB is lower due to the reduced number of training examples at this level. The effect is more pronounced with the MES expert as there is a higher degree of variability in the MES data, compared to the acoustic data.

The Borda count methods have classification accuracies equal or better than the acoustic expert at all noise levels, except at

[3]The Lombard effect can be simply described as a person's tendency to speak louder in an acoustically noisy environment [27].

0-dB (e_A 97.67%; E_{BA} 91.00%; EBM 89.67%). The decrease in classification accuracy at the 0-dB noise level is due to the decreased performance of the MES expert at this level causing a large difference in accuracy between the MES and acoustic expert. Although the Borda count methods have classification accuracies higher than the acoustic expert, at noise levels 12-dB or greater, the accuracies are significantly lower than the MES expert. When the acoustic expert is favored in the Borda count method during ties (E_{BA}), superior performance is shown at the lower noise levels, which is expected as the acoustic expert outperforms the MES expert at these noise levels. Similarly, when the MES expert is favored in the Borda count method during ties (E_{BM}), superior performance is shown at the high noise levels, where the MES expert outperforms the acoustic expert.

The score-based method outperforms either Borda count method (E_{BA} or E_{BM}) in all cases, except at the 12-dB noise level (E_{BM} 65.50%; E_S 64.00%). Similar to the Borda count methods, the score-based method outperforms the acoustic expert at all noise levels, except at 0-dB (e_A 97.67%; E_S 92.83%). As the acoustic expert degrades in performance at high noise levels, the score-based method also degrades in performance with classification accuracies much lower than the MES expert.

A noticeable difference can be seen when the opinion of experts were combined using the plausibility method. As with the other combination methods, classification accuracies were all higher than the acoustic expert, except at the 0-dB noise level (e_A 97.67%; E_P 94.83%). Unlike, the other methods, as the noise level increases to 12 dB and beyond, the classification accuracy of the plausibility method did not degrade along with the acoustic expert; instead, the classification accuracy follows the accuracy of the MES expert. This result is desirable since at high noise levels, the acoustic expert is essentially randomly guessing and, therefore, not providing any useful information. Thus, the highest possible classification accuracy should be equal to the classification accuracy of the MES expert.

At the 6- and 9-dB noise level, while the classification accuracy of the plausibility method exceeds the acoustic expert, it is lower than the Borda count methods and score-based method by as much as 3.75%.

A. Incorporating a priori Information

Thus far, the plausibility method of combining the opinion of experts was applied with no *a priori* knowledge, treating each expert equally; however, one could easily incorporate *a priori* knowledge by changing the mean of the cumulative distribution function G_k for expert e_k adjusting the overall confidence in that expert. Under noise free conditions, the MES expert is not as reliable as the acoustic expert. Indeed, the MES expert's highest classification accuracy is only around 90%. Changing the mean of the cumulative distribution function for the MES expert from $2\sigma_k$ to $3\sigma_k$, causes a larger portion of belief to be assigned to the whole set Θ (vacuous belief), reflecting a lower overall reliability of the MES expert.

Fig. 4 plots the classification accuracy of this modified plausibility method (E_{P2}), along with the original plausibility method (E_P) and the score-based method (E_S) for comparison. The Borda count methods were not included as the score-based method outperforms these methods. Incorporating the *a priori* knowledge of the lower overall reliability of the MES expert results in an increased classification accuracies at low noise levels (E_{P2} 98.67% at 0 dB), with a tradeoff of decreased classification accuracies at high noise levels.

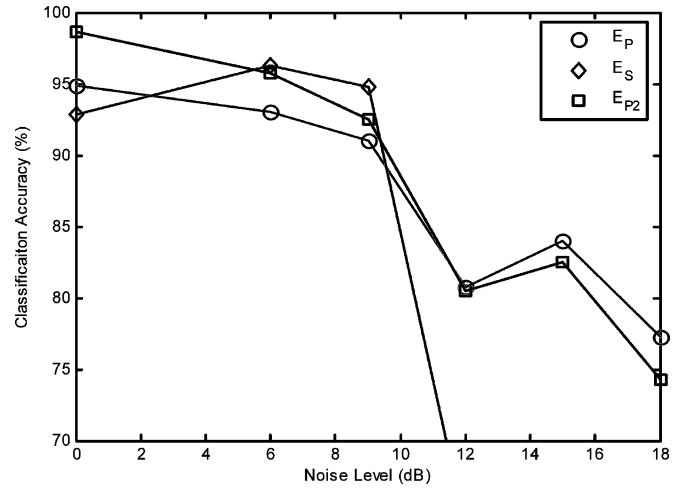

Fig. 4. Classification accuracies for score-based method (E_S), and plausibility method (E_P), and plausibility method compensating for overall lower reliability of MES expert (E_{P2}).

B. Incorporating a Third Expert

Benefits of using multiple experts for multiple sources of information have been demonstrated with increased classification accuracy and system robustness. It can also be beneficial to use multiple experts on a single channel of information. Different classification methodologies have been shown to have different strengths and weaknesses. An artificial neural network, for example, is an excellent classifier for discerning classes that have a complex nonlinear separation surface; however, it does not deal with time-scale and shape variance as well as other classifiers, such as HMMs [28]. The use of multiple classifiers on a single channel of information has already demonstrated improved classification rates in handwriting recognition [29], [30], person identification [4], [16], [19], and lip reading [20].

We chose to include the MES classification methodology by Englehart *et al.* [6] to serve as a third expert (e_{M2}). This method was previously used in [11] for MES ASR, where it was found that the best performance was obtained extracting features based on the wavelet transform. Using principal component analysis for feature reduction, a reduced set of 24 features was presented to the classifier. The method by Englehart *et al.* [6] suggests the usage of a linear discriminant analysis classifier; however, a linear perceptron was used instead because it could produce an output that mimicked confidence probabilities, which were used as the expert's measurement vector. The linear perceptron had 10 outputs, with an output corresponding to each class, and was trained to output a one for the true class and a zero for the remaining classes. Classification accuracies obtained for the linear perceptron classifier were equal to the classification accuracies obtained using linear discriminant analysis.

The overall reliability of this new MES expert e_{M2} was expected to be lower than expert e_M, which uses HMMs to classify the MES. This knowledge was accommodated in the plausibility

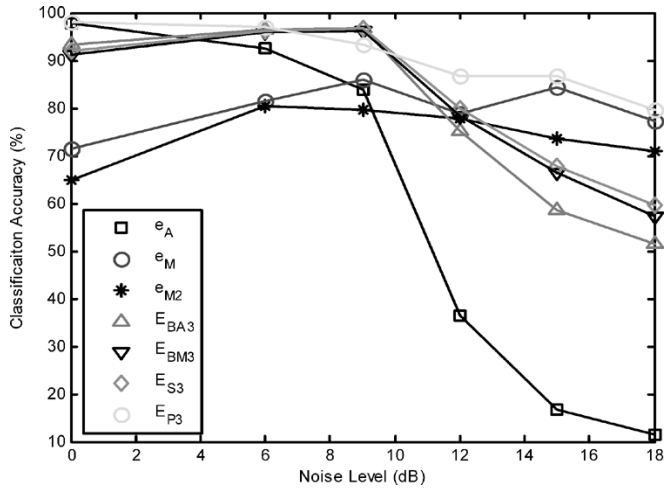

Fig. 5. Classification accuracies for individual experts: acoustic expert $(\mathrm{e_A})$, HMM MES expert $(\mathrm{e_M})$, linear perceptron MES expert $(\mathrm{e_{M2}})$; and three-expert systems: Borda method favoring acoustic expert $(\mathrm{E_{BA3}})$, Borda method favoring MES expert $(\mathrm{E_{BA3}})$, score-based method $(\mathrm{E_{S3}})$, and plausibility method $(\mathrm{E_P})$.

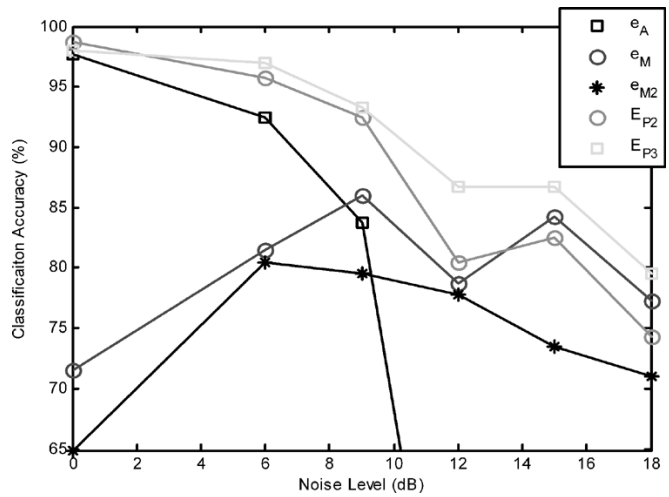

Fig. 6. Classification accuracies for acoustic expert $(\mathrm{e_A})$, the MES expert using HMMs $(\mathrm{e_M})$, the MES expert using a linear perceptron $(\mathrm{e_{M2}})$, the plausibility method with two experts $(\mathrm{E_{P2}})$, and plausibility method with three experts $(\mathrm{E_{P3}})$.

method of combining the opinion of experts $(\mathrm{E_{P3}})$ using a mean of $4\sigma_\mathrm{k}$ for the cumulative distribution function for the $\mathrm{e_{M2}}$ expert.

Fig. 5 plots the classification accuracies of the individual experts, as well as the classification accuracies for the three-expert systems, using the Borda count method, score-based method, and plausibility method. As expected, expert $\mathrm{e_{M2}}$ has a lower classification accuracy than expert $\mathrm{e_M}$ (5.36% difference on average). Classification accuracies of the three-expert systems, for all combination methods, were higher than their corresponding two-expert system (Fig. 3). Similar characteristics that were found in the two-expert systems are also apparent in the three-expert systems. The Borda count and score-based based methods all provided classification accuracies that exceed the acoustic expert, except at the 0-dB level; however, as the acoustic expert degraded at high noise levels, the classification accuracy of these combination methods fell significantly lower than the classification accuracies of both MES experts; however, the degradation in performance was not as severe as it was in the two expert system.

Fig. 6 is a plot of the two- and three- expert classification accuracies using the plausibility method. Similar to the two-expert system the plausibility method did not degrade along with the acoustic expert in the three-expert system. In fact, this three-expert system was the only combination method to maintain classification accuracies higher than any individual expert at all noise levels. It also provided the highest classification accuracy among all methodologies at all noise levels except at 9-dB ($\mathrm{E_{P3}}$ 93.25%; $\mathrm{E_{BA}}$ and $\mathrm{E_S}$ 96.75%).

VI. Discussion

Additional speech information provided by the MES has been successfully used to augment a conventional acoustic ASR expert. Improvements in classification accuracies were demonstrated for a large range of acoustic noise levels. Traditional methods of combination (i.e., Borda count and score-based methods) were successful in improving the classification accuracy of the acoustic ASR expert; however, as the reliability of the acoustic expert deteriorated with increasing noise, so did the performance of these methods of combination. These traditional methods of combination were unsuccessful at utilizing the MES expert's resilience to acoustic noise, resulting in classification accuracies much lower than the MES expert.

Unlike these other methods of combination, the plausibility method was able to track the reliability of the acoustic expert dynamically. This multiexpert system was able to produce classification accuracies that exceeded the accuracies of the individual experts at all noise levels. Note that this was accomplished without any external monitoring of the environment or training at multiple noise levels (i.e., the plausibility method did not rely on a measure of the level of acoustic noise). The reliability of the expert was simply determined by the interpretation of the scores in relationship to the training scores through a nonlinear mapping of the score differences into a BPA. Training only at the 0-dB noise level, the plausibility method was capable of maintaining classification accuracies above 78.75% across an 18-dB range of acoustic noise.

While it may be possible to obtain similar classification accuracies from an individual expert, using methods such as noise cancellation techniques and noise resilient features, it is doubtful that the system will be generalizable to other noise conditions (e.g., colored noise, vibration, speaker stress). In addition, any improvement in the performance of the individual experts is expected to result in improved performance in the multiexpert system. The generalizability of the multiexpert system, using the plausibility method, also eliminates the need for training examples in multiple noise environments, which is both cumbersome and often not desirable.

It was also demonstrated that improved performance can also be achieved using multiple classification methodologies on a single channel of information. The addition of an expert, whether it uses an additional source of information or a different pattern recognition methodology, can improve a multiexpert

system as long as it provides information that is not redundant. In evidence theory, proper application of Dempster's rule of combination requires that the two bodies of evidence that are to be combined exhibit some type of independence. Dempster [21] refers to this independence as "independent sources of information" and Shafer [22] refers to this as "distinct body of evidence". Within evidence theory, independence cannot be simply equated with stochastical independence [31]. Also, proposed independence requirements for the use of Dempster's rule of combination have been questionable, unclear, or vague. Although, the independence of the experts is difficult to prove with mathematical rigor, the concept of providing nonredundant information can be understood intuitively. This point can be understood by imagining two-expert system, in which both experts are identical implementations. While correct classifications would be made with apparent confidence (i.e., both experts agree), misclassification would also be reinforced.

VII. Conclusion

A pattern recognition system can be enhanced using a multiexpert system. Individual experts in the system may operate on different sources of information or may utilize the same source of information in differing ways. If the opinions of the individual experts are not redundant, they can be combined to form a multiexpert decision that takes advantage of the strengths of the individual experts, while diminishing their weaknesses. The advantages of a multiexpert system are dependent on the method of combining the opinion of experts.

In this paper, a multiexpert ASR system was successfully implemented using an acoustic ASR expert and a MES expert. Although traditional methods (i.e., Borda count and score-based methods) of combining the opinion of experts improved the classification accuracies of the individual experts, their performance was not satisfactory. In particular, these methods did not cope with acoustic expert as it failed with increasing noise. The new plausibility method presented in this paper, however, manages to track the reliability of the individual experts and was able to maintain relatively high classification accuracies across a wide range of acoustic noise.

Continued testing and development of the plausibility method will hopefully prove the generalizability of this method, by demonstrating its utility for ASR in other noise environments (e.g., colored noise, vibration) and for other pattern recognition applications (e.g., handwriting recognition, person identification, myoelectrically controlled prostheses). This benefits of the multiexpert ASR system may also be useful in other application environments, including construction sites, for firefighters, and for persons with speech impairments. Within the application of ASR in high performance jet aircraft, the effects of other sources of noise, such as vibration, positive pressure breathing, high G-force, also need to be investigated as these would likely have a large effect on the MES ASR expert.

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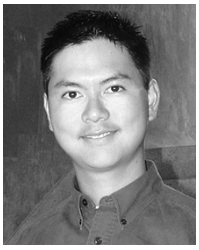

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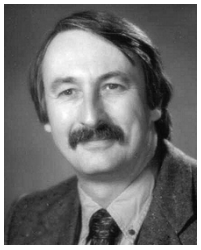

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